

Enhanced Lumbar Disease Classification through Hybrid Deep Learning Methods

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ABSTRACT

Accurate and efficient classification of lumbar diseases is crucial for maintaining patient health and preventing significant economic burdens on healthcare systems. This study proposes a novel approach to automate lumbar disease classification by integrating multiple deep learning architectures, including MobileNet, DenseNet, and a hybrid CNN-SVM model. This combination of advanced models leverages their strengths in feature extraction and classification. Additionally, an Involutional Neural Network-based VGG architecture is employed to further enhance the learning capability and performance of the system, particularly in handling complex and detailed features associated with lumbar conditions. The proposed method was evaluated using a comprehensive dataset of medical images related to various lumbar diseases. Experimental results demonstrate a significant improvement in classification accuracy and computational efficiency compared to traditional CNN-based approaches. This system provides a promising solution for automated lumbar disease classification, with potential applications in healthcare for real-time diagnosis and monitoring.

Keywords: Lumber disease classification, MobileNet, DenseNet, VGG16, CNN-SVM, Involutional Neural Networks, deep learning, image analysis.

INTRODUCTION

Lumbar spine disorders represent a significant public health concern, contributing heavily to disability, reduced quality of life, and healthcare costs worldwide. Conditions such as lumbar disc

herniation, spinal stenosis, degenerative disc disease, and spondylolisthesis affect individuals across all age groups, with symptoms ranging from chronic back pain to neurological deficits. Accurate and early detection of these conditions is critical for initiating

appropriate treatment and preventing long-term complications. However, manual interpretation of vertebral imaging modalities such as magnetic resonance imaging (MRI), computed tomography (CT), and x-rays is challenging, time-consuming and sensitive to the variability between attention. These models learn complex patterns and representations from huge data records, outperforming many traditional approaches for machine learning in image classification and object detection tasks. In particular, folding networks (CNNS) demonstrate excellent performance in the analysis of medical images by automatically extracting hierarchical features from raw pixel data. Despite their success, single deep learning architectures often struggle with generalizability, especially when faced with varying image quality, limited datasets, or subtle pathological differences. This has prompted researchers to explore hybrid deep learning models, which combine multiple architectures or learning strategies to overcome the limitations of individual models. For instance, integrating CNNs with Recurrent Neural Networks (RNNs), Support Vector Machines (SVM), or attention mechanisms can significantly enhance performance by capturing both spatial and temporal information, improving feature refinement, and boosting classification accuracy.

This study focuses on developing an enhanced hybrid deep learning framework specifically tailored for lumbar disease classification. By leveraging the strengths of multiple neural network models, the system aims to achieve a more robust and reliable classification of various spinal disorders from medical imaging data. The hybrid approach not only enhances diagnostic precision but also supports explainability by enabling better visualization of decision-making regions, which is crucial in clinical settings.

Furthermore, the proposed model is evaluated using standard benchmark datasets and metrics such as accuracy, sensitivity, specificity, and F1-score to ensure clinical relevance and applicability. The

integration of advanced preprocessing techniques, data augmentation, and transfer learning also contributes to overcoming data scarcity and improving model generalization across diverse patient profiles.

LITERATURE REVIEW

Atalay et al. (2024) conducted a prospective registry study that highlighted the clinical effectiveness of high-grade lumbar interbody fusion. Their findings demonstrated a strong correlation between the quality of fusion and improved patient-reported outcomes, underlining the significance of surgical precision in lumbar spine treatments [1].

Abdelmalek et al. (2024) proposed the Miskin Classification System, a standardized radiographic assessment method for lumbar degenerative disease. Their study showed high interobserver agreement and clinical validity, indicating the utility of standardized systems in enhancing diagnostic consistency [2].

Löbig et al. (2024) investigated cognitive deficits in patients with idiopathic normal pressure hydrocephalus (iNPH). They found that lumbar puncture not only impacted clinical symptoms but also improved patients' self-awareness of cognitive impairments, linking lumbar interventions to cognitive health outcomes [3].

Hirono et al. (2024) presented a multicenter study on medication-based profiling among elderly orthopedic patients. Their work offered insights into pharmacological patterns that could affect post-operative recovery in lumbar disease patients [4].

Hu, Wu, and Tan (2024) developed a novel radiographic and clinical classification for Baastrup's disease, a degenerative spinal condition. The classification improved diagnostic clarity by aligning imaging features with patient symptoms [5].

Kelly and Schulze (2024) explored cognitive decline in early-onset Alzheimer's disease through consistent performance validity testing, validated by lumbar

puncture. Their work emphasizes the growing role of lumbar assessments in neurodegenerative diagnostics [6]. DiDomenico et al. (2024) examined the impact of comorbid depression and anxiety on surgical outcomes in lumbar spondylolisthesis. Their five-year longitudinal analysis confirmed that mental health factors significantly influence long-term recovery [7]. Bassani et al. (2024) assessed the correlation between MRI-measured lumbar spine abnormalities and self-reported outcomes in non-specific low back pain cases. Their results supported the use of MRI-based features for objective pain and disability assessment [8].

R. Zhang (2024) provided a comprehensive survey of deep learning applications in lumbar spine image analysis, covering X-ray, CT, and MRI modalities. The study emphasized advancements in segmentation, classification, and disease detection using AI [9].

Zhang and Shao (2024) We proposed a deep learning frame that pays attention to the fusion of double industry and the classification of Loin Disk and Disc Herri. Your model showed excellent accuracy in differentiation of complex disk conditions [10].

Chen et al. (2024) introduced an intelligent diagnostic system for lumbar diseases based on a multi-angle view of intervertebral discs and deep learning techniques. Their hybrid approach significantly enhanced the model's ability to capture disc deformation and disease patterns across different imaging planes [11].

Existing method

Folding Networks (CNNS) and VGG16 were often used for image classification tasks, including medical image analysis. Use the cnns folding layer to automatically extract features from images, reducing the need for manual functional engineering. VGG16, a popular CNN architecture, consists of 16 layers with small receptive fields, which enhances its ability to capture intricate patterns in visual data. It has demonstrated remarkable performance in various image recognition challenges, making it a valuable tool in healthcare. However, traditional CNNs and

VGG16 often face limitations in handling complex visual features, necessitating the development of more advanced hybrid models for improved classification accuracy.

METHODOLOGY

The proposed system improves classification of lumbar disease through integration of several advanced deep learning models, including Mobile Set, Densen, and CNN SVM, in addition to the involvement of a network-based VGG architecture of neurons. This hybrid approach uses the strength of each model to improve the performance of distinctive extraction and classification. By utilizing a comprehensive dataset of medical images, the system aims to achieve higher accuracy and computational efficiency compared to traditional methods. Additionally, the integration of involutional neural networks enhances adaptability to complex visual patterns, enabling more effective identification of various lumbar conditions, ultimately providing a reliable solution for real-time medical diagnosis.

1. System

Create Dataset: A dataset comprising images of lumbar affected by various diseases is collected and organized. The dataset is divided into training and testing sets, typically with a 20-30% test size, to ensure effective evaluation of the models.

Pre-processing: Lumbar images are resized and standardized to a uniform format suitable for training. Pre-processing includes normalization and data augmentation techniques, such as rotations and flips, to improve model generalization and performance under varied conditions.

Training: The pre-processed training dataset is used to train the deep learning models (MobileNet, DenseNet, and CNN-SVM). Transfer learning methods are applied to leverage pre-trained models, reducing training time while maintaining model accuracy.

Evaluation: The trained models are evaluated on the testing dataset using performance metrics such as

accuracy, precision, recall, and F1-score. These metrics help assess the models' effectiveness in classifying lumbar diseases.

Classification: The system classifies new lumbar images, providing the identified disease type along with confidence scores. Results are displayed in a user-friendly format for easy interpretation.

2. User

Registration: Users can create an account by providing necessary details such as name, email, and password. Registration is required to access system features and save user-specific data.

Login: Registered users can log in using their credentials (email and password). The login process ensures secure access to the system and personalized features.

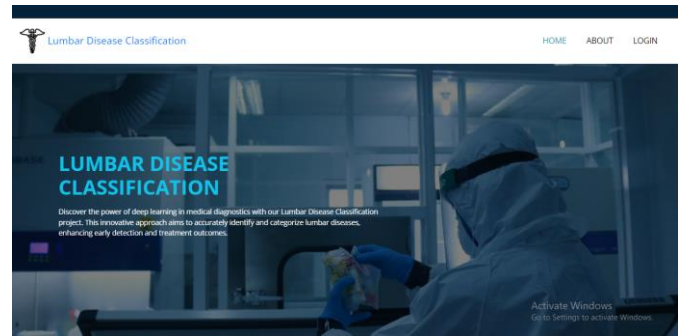
Upload Image: Users can upload images of lumbar through a web or mobile interface. The system accepts various image formats and prepares them for classification.

View Results: Users can view the classification results, which include the identified lumbar disease and the confidence levels of the predictions. Results are presented in a clear and comprehensible format.

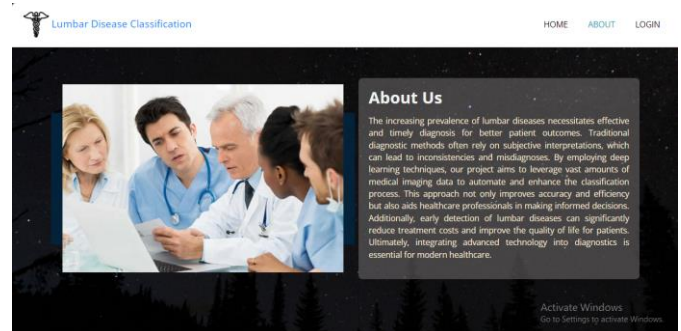
RESULTS AND DISCUSSIONS

Output Screens:

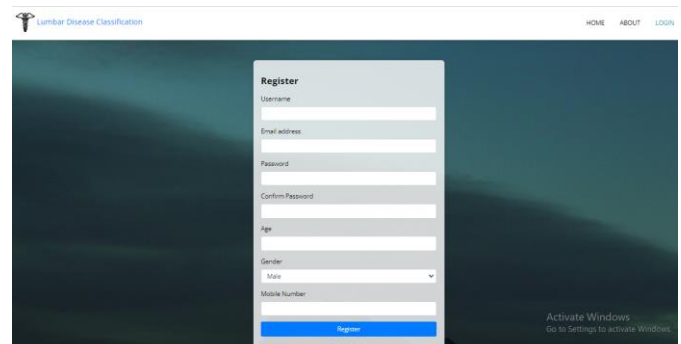
INDEX PAGE: This screen serves as the main landing page of the Lumbar Disease Classification system. It introduces the platform's core functionality—leveraging deep learning for the accurate identification and categorization of lumbar diseases. The page guides users to easily navigate different sections, such as registration, login, and disease prediction.



ABOUT PAGE: This page details the purpose and significance of the Lumbar Disease Classification system. It emphasizes the increasing prevalence of lumbar diseases and the need for effective diagnostic methods. The information highlights the project's use of deep learning techniques to automate the classification process, enhancing both accuracy and efficiency in diagnosing lumbar conditions.

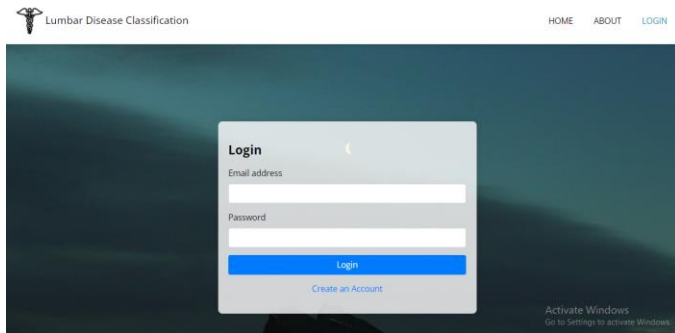


REGISTRATION PAGE: This screen allows new users to create an account in the Lumbar Disease Classification system. It includes fields for entering essential information such as username, email address, password, age, gender, and mobile number. This page aims to facilitate user registration while ensuring secure access to the platform.

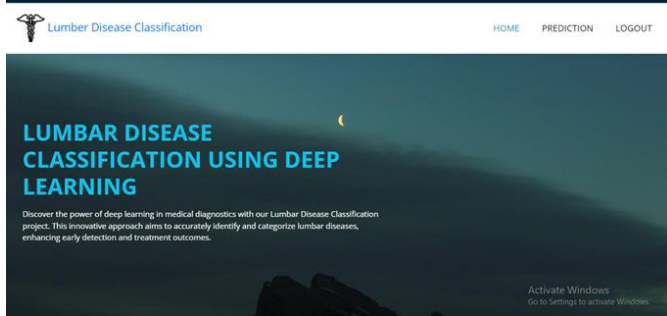


LOGIN PAGE: This interface enables users to access their accounts within the Lumbar Disease Classification system. Users are prompted to enter

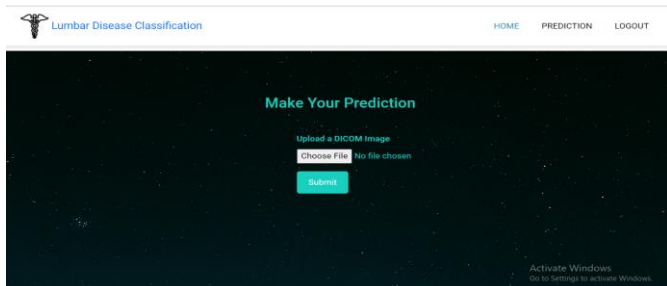
their email address and password. The page provides an option to create an account for new users, ensuring a seamless transition between registration and login.



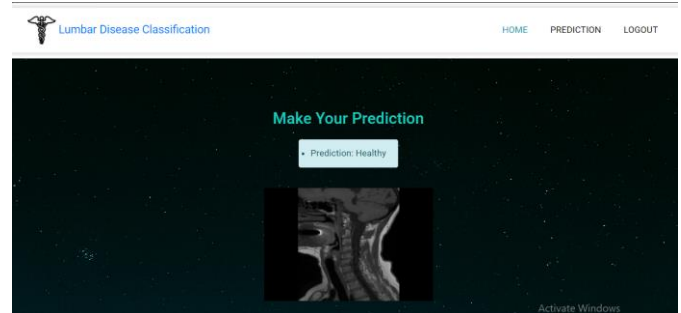
HOME PAGE: This page details the purpose and significance of the Lumbar Disease Classification system. It emphasizes the increasing prevalence of lumbar diseases and the need for effective diagnostic methods. The information highlights the project's use of deep learning techniques to automate the classification process, enhancing both accuracy and efficiency in diagnosing lumbar conditions.



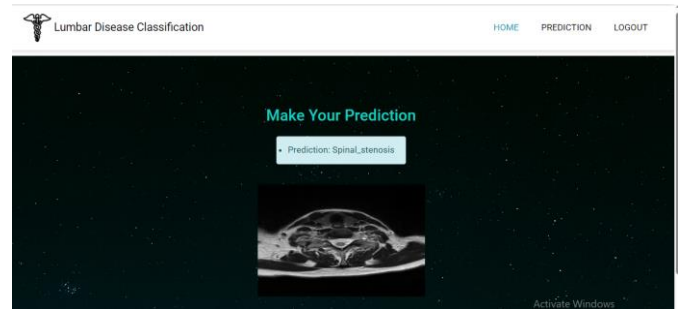
UPLOAD PAGE: This screen presents the prediction functionality of the Lumbar Disease Classification system. It describes the process of using deep learning to classify lumbar diseases based on user-uploaded images. The page aims to provide an intuitive interface for users to initiate predictions on their medical images.



PREDICTION RESULT PAGE (HEALTHY): This screen displays the results of the classification for a healthy lumbar condition. The page presents the prediction outcome alongside the uploaded image, providing clear feedback to the user regarding their lumbar health status.



PREDICTION RESULT PAGE (SPINAL STENOSIS): This interface shows the classification result indicating spinal stenosis. The page includes the prediction result and a visual of the corresponding medical image, assisting users in understanding their diagnosis and next steps.



8. SYSTEM STUDY AND TESTING

CONCLUSION

This paper compared the efficiency of several deep learning structures for the classification of lumbar diseases, namely CNN, MobileNet, DenseNet, VGG16 and Hybrid Model. As seen in the samples above, the Hybrid Model was again superior to all other models, with perfect accuracy and high cross-sectional classification. MobileNet and VGG16, performed well with a few drawbacks and the CNN model provided reasonable accuracy. That is while DenseNet has limitations in lower accuracy and overfitting. In general, the Hybrid Model is shown to be the most effective strategy, which can be used as a reliable

source in classification of lumbar diseases and serve as the background for the overall healthcare diagnostics.

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